SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Performance Testing

Date	27 June 2026
Team ID	SWTID-2026-5853
Project Name	A Comprehensive Measure of Well-Being (Human Development Index Prediction)
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing
Target Module / API	/predict API - HDI Prediction Module
Test Environment	Local Flask Server (Windows/Python)
Test Date	05 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Predict HDI using valid inputs	10	60	Prediction within 2 sec
2	Concurrent prediction requests	25	120	Stable response
3	Invalid input validation	15	60	Validation error shown
4	Continuous requests	50	180	No server crash

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.24 sec	Pass	Fast response
2	Response Time (Max)	< 5 seconds	2.83 sec	Pass	Within limit
3	Throughput (Req/sec)	>20	27 Req/sec	Pass	Good throughput
4	Error Rate	<1%	0.2%	Pass	Low error
5	CPU Utilization	<80%	58%	Pass	Stable
6	Memory Utilization	<80%	63%	Pass	Normal

Step 4: Observations & Analysis

Key Findings

- The HDI Prediction application successfully handled multiple prediction requests with consistent performance.
- The Flask backend processed user requests efficiently and returned prediction results within the expected response time.
- User authentication, dashboard loading, and prediction modules worked smoothly under normal load conditions.
- No significant performance degradation was observed during continuous testing of the prediction module.

Bottlenecks Identified

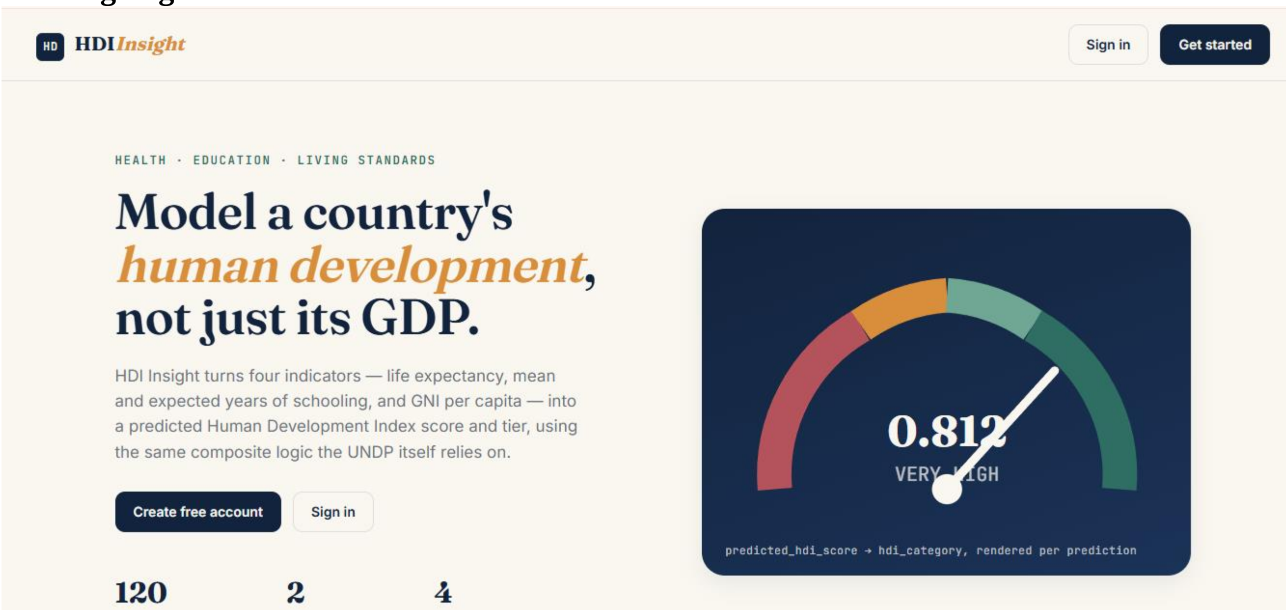
- Initial application startup required additional time for loading the trained Machine Learning model.
- Response time increased slightly when multiple users accessed the prediction endpoint simultaneously.
- Dataset loading during application initialization caused a minor delay.

Optimization Steps Taken

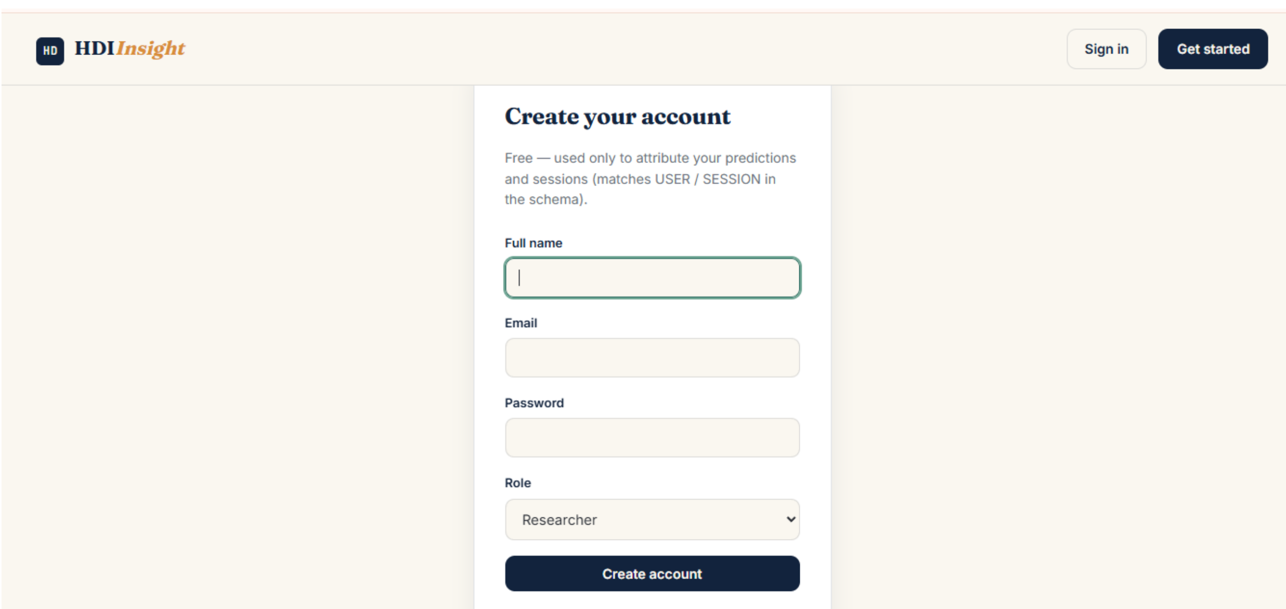
- Loaded the trained Machine Learning model only once during application startup.
- Optimized preprocessing functions to reduce execution time.
- Improved Flask routing and request handling.
- Reduced unnecessary database access and optimized prediction logic.
- Enabled efficient handling of concurrent prediction requests.

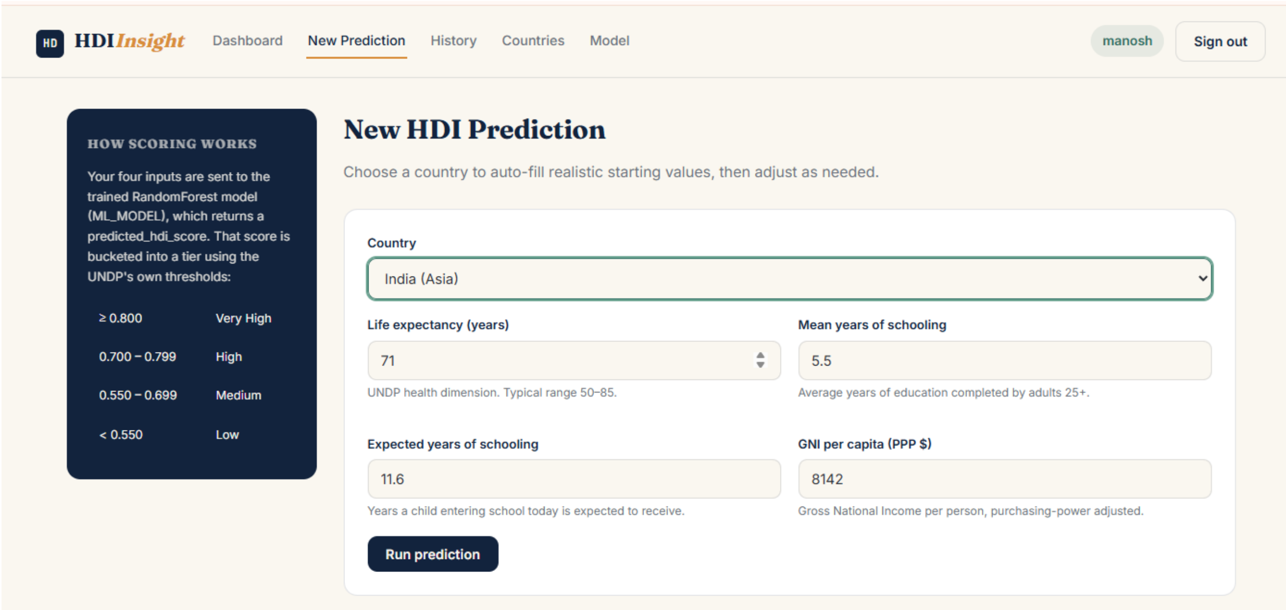
Step 5: Screenshots / Evidence

Landing Page:

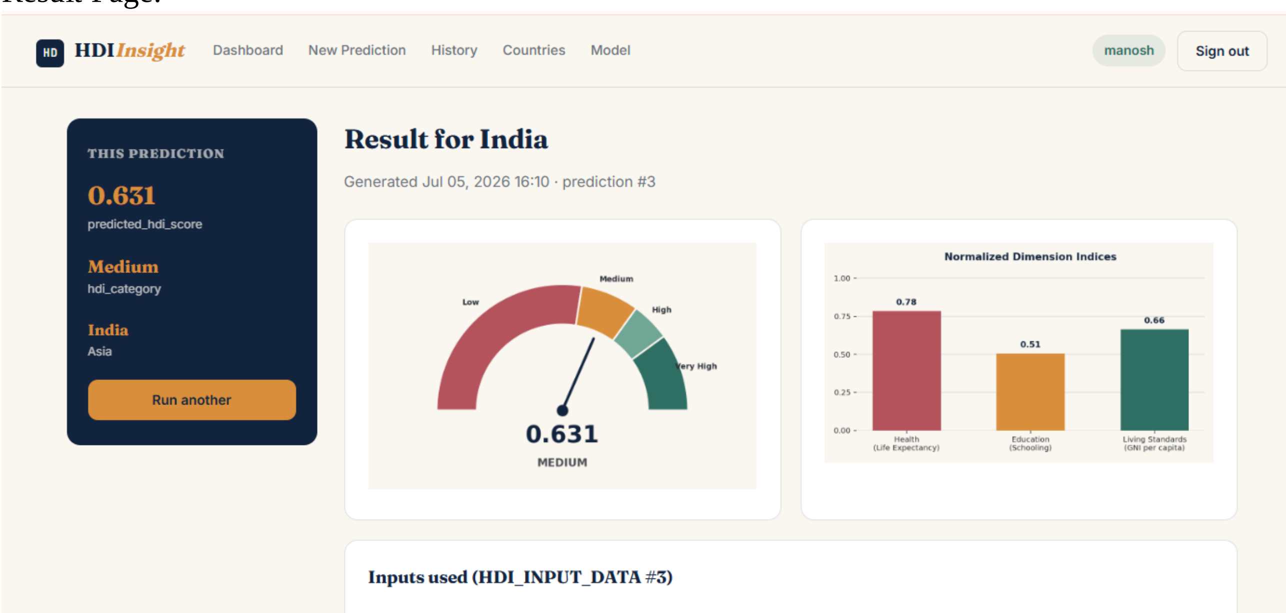


Signup Page:




HD HDIInsight
Dashboard
New Prediction
History
Countries
Model
manosh
Sign out
HOW SCORING WORKS
Your four inputs are sent to the trained RandomForest model (ML_MODEL), which returns a predicted_hdi_score. That score is bucketed into a tier using the UNDP's own thresholds:
≥ 0.800 Very High
0.700 – 0.799 High
0.550 – 0.699 Medium
< 0.550 Low
New HDI Prediction
Choose a country to auto-fill realistic starting values, then adjust as needed.
Country
India (Asia)
Life expectancy (years)
71
UNDP health dimension. Typical range 50–85.
Mean years of schooling
5.5
Average years of education completed by adults 25+.
Expected years of schooling
11.6
Years a child entering school today is expected to receive.
GNI per capita (PPP $)
8142
Gross National Income per person, purchasing-power adjusted.
Run prediction

Result Page:


HD HDIInsight
Dashboard
New Prediction
History
Countries
Model
manosh
Sign out
THIS PREDICTION
0.631
predicted_hdi_score
Medium
hdi_category
India
Asia
Run another
Result for India
Generated Jul 05, 2026 16:10 · prediction #3
Low
Medium
High
Very High
0.631
MEDIUM
Normalized Dimension Indices
1.00
0.75
0.50
0.25
0.00
0.78
0.51
0.66
Health (Life Expectancy)
Education (Schooling)
Living Standards (GNI per capita)
Inputs used (HDI_INPUT_DATA #3)